

Martin van Beek

PERSONAL INFORMATION

ADDRESS: Room 1.218, Alan Turing Building,
University of Manchester, Oxford Road,
Manchester, M13 9PL

WORK EMAIL: martin.vanbeek@manchester.ac.uk

PERSONAL WEBSITE: martinvanbeek.github.io

EDUCATION

SEPTEMBER 2017 - JULY 2022 PhD in Pure Mathematics
DEFENDED DECEMBER 2021 University of Birmingham
Thesis: Local Group Theory, The Amalgam Method, and
Fusion Systems
Supervisor: Prof. Chris Parker

OCTOBER 2016 - JUNE 2017 Master of Advanced Study in Mathematical Sciences
University of Warwick
Classification: Masters with Distinction
Thesis: Properties of the 7-Sphere
Supervisor: Prof. Dmitriy Rumynin

SEPTEMBER 2012 - JULY 2016 Bachelors of Science in Mathematics
University of Glasgow
Classification: Honours of the First Class
Thesis: The Large Mathieu Groups
Supervisor: Dr. Andrew Baker
Awards: Dougall Prize 2015

RESEARCH INTERESTS

I am currently interested in local group theory, specifically through the lens of saturated fusion systems over p -groups and their interactions with group amalgams. In fusion systems, my current research focuses on recognizing the p -fusion categories of simple groups of Lie type from prescribed local actions. Alongside this, I am interested in exotic fusion systems at odd primes, and the applications of fusion systems to representation theory and homotopy theory.

EMPLOYMENT

OCTOBER 2022 - HEILBRONN FELLOW AT THE UNIVERSITY OF MANCHESTER
PRESENT

The aim of this research is two-fold. Firstly, to work towards recognizing the p -fusion category of the groups of Lie type via the construction and recognition of parabolic systems in fusion systems, and ultimately to classify large families of fusion systems of parabolic characteristic p . Secondly, to uncover exotic fusion systems on p -groups, for p an odd prime, with the goal of identifying distinguishing features of exotic fusion systems when compared to p -fusion categories of finite groups.

NOVEMBER 2021 - RESEARCH ASSOCIATE AT THE UNIVERSITY OF BIRMINGHAM
OCTOBER 2022

I was funded by the EPSRC for 12 months to facilitate research pertaining to saturated fusion systems. During this time, I discovered new exotic fusion systems which are related to sporadic simple groups, completely determined a large class of fusion systems supported on certain Sylow p -subgroups of a simple group of Lie type of rank two, and began the groundwork on describing fusion systems which contain a parabolic system.

PUBLICATIONS AND PREPRINTS

Books:

1. Rank 2 amalgams and fusion systems, ~ 200 pp., ([Arxiv](#)), *Lecture Notes in Mathematics*, Springer, (2024)

Published Articles:

1. Groups with conjugacy classes of coprime sizes, with Rachel D. Camina, Attila Maróti, Emanuele Pacifici, Chris Parker, Kamilla Rekvényi, Jack Saunders, Víctor Sotomayor & Gareth Tracey, ([Arxiv](#)), (2026)
2. Fusion systems related to polynomial representations of $\mathrm{SL}_2(q)$, with Valentina Grazian, Chris Parker & Jason Semeraro, ([Arxiv](#)), *Journal of the LMS*, (2026)
3. Lifting polynomial representations of $\mathrm{SL}_2(q)$ from $\mathbb{F}_p$ to $\mathbb{Z}/p^s\mathbb{Z}$, with Chris Parker, ([Arxiv](#)), *Proceedings of the AMS*, (2026)
4. Exotic fusion systems related to sporadic simple groups, ([Arxiv](#)), *Math. Proc. Camb. Phil. Soc.*, (2025)
5. Fusion systems on Sylow p -subgroups of rank 2 simple groups of Lie type, ([Arxiv](#)), *Forum of Mathematics, Sigma*, (2024)
6. Fusion systems on a Sylow p -subgroup of $G_2(p^n)$ or $\mathrm{PSU}_4(p^n)$, ([Arxiv](#)), *Journal of Algebra*, (2023)

Submitted Articles:

1. Fusion systems and simple groups with class two Sylow p -subgroups, ([Arxiv](#)), *Submitted*

In Preparation:

1. Small modules for groups with a strongly p -embedded subgroup, with Valentina Grazian, Justin Lynd, Bob Oliver, Chris Parker & Jason Semeraro
2. Fusion systems with a normal, abelian, essential subgroup I, with Valentina Grazian, Justin Lynd, Bob Oliver, Chris Parker & Jason Semeraro
3. Fusion systems with a soft subgroup, with Valentina Grazian & Chris Parker
4. Fusion systems on 3-groups of sectional rank 3, with Valentina Grazian & Chris Parker
5. Fusion systems with a normal, abelian, essential subgroup II, with Valentina Grazian, Justin Lynd, Bob Oliver, Chris Parker & Jason Semeraro

CONFERENCES AND SEMINARS (CO-)ORGANIZED

Patterns in exotic fusion systems (Funded by the Heilbronn Institute)	- Univ. of Birmingham	- May 2023
Postgraduate group theory conference (Funded by the LMS and Heilbronn Institute)	- Univ. of Birmingham	- Jul. 2019

FUNDING

Heilbronn focused research grant (Joint with Prof. Chris Parker)	- £6000	- Mar. 2023
---	---------	-------------

SELECTED TALKS

Algebra Seminar <i>“Realizing Representations in Finite Groups and Fusion Systems”</i>	- Univ. di Padova	- Apr. 2025
Algebra Seminar <i>“Fusion systems and simple groups with class two Sylow p-subgroups”</i>	- Univ. of Birmingham	- Feb. 2025
Norddeutsches Gruppentheorie-Kolloquium <i>“Fusion systems and simple groups with class two Sylow p-subgroups”</i>	- TU Dresden	- Nov. 2024
Algebra Seminar <i>“Exotic fusion systems related to sporadic simple groups”</i>	- Univ. of Warwick	- Nov. 2023
Algebra Seminar <i>“Saturated fusion systems and rank two simple groups of Lie type”</i>	- Univ. of Louisiana	- Feb. 2023
Groups and Topological Groups Conference	- Univ. Halle-Wittenberg	- Feb. 2023

SELECTED TEACHING & SUPERVISION

2025 - Present CO-SUPERVISE A PhD CANDIDATE AT THE UNIVERSITY OF MANCHESTER

My responsibilities include suggesting a project, guiding the student towards to solutions via weekly meetings, and recommending relevant literature for the project and more general growth.

2024 LECTURED “TOPICS IN FINITE p -GROUPS” AT THE UNIVERSITY OF MANCHESTER

My responsibilities included devising the course, writing lecture notes, administering the lectures and problem sessions, and writing and marking the final exam. The audience was a mix between PhD students and postdocs.

SERVICE

I have refereed articles for LMS journals, the Journal of Algebra and for the Journal of Group Theory.